Hojun Lee

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EDUCATION

- **Purdue University – West Lafayette, IN, USA** **2023 – Present**
  Ph.D. in Mechanical Engineering
- **University of Michigan – Ann Arbor, MI, USA** **2020 – 2022**
  M.S. in Robotics
- **Purdue University – West Lafayette, IN, USA** **2014 – 2020**
  B.S.E. in Mechanical Engineering

RESEARCH EXPERIENCE

- **School of Mechanical Engineering, Purdue University – West Lafayette, IN, USA** 8/2023 – Present
  Graduate Research Assistant
- **Department of Mechanical Engineering, University of Michigan – Ann Arbor, MI, USA** 6/2022 – 5/2023
  Research Associate
- **Robotics Department, University of Michigan – Ann Arbor, MI, USA** 1/2021 – 5/2022
  Graduate Research Assistant
- **School of Mechanical Engineering, Purdue University – West Lafayette, IN, USA** 5/2019 – 5/2020
  Undergraduate Research Assistant
- **Polytechnic Institute, Purdue University – West Lafayette, IN, USA** 5/2015 – 5/2016
  Undergraduate Research Assistant

TEACHING EXPERIENCE

- **Purdue University – West Lafayette, IN, USA** 7/2026 – 8/2026
  Teaching Assistant: Instructed ML tutorials for exchange students from South Korea "2026 KIAT-Purdue Summer Exchange Program"
- **Purdue University – West Lafayette, IN, USA** 1/2026 – 5/2026
  Teaching Assistant: Instructed ML tutorials for graduate students (ME597) "Industrial IoT Implementation for Smart Manufacturing"
- **Purdue University – West Lafayette, IN, USA** 1/2025 – 5/2025
  Teaching Assistant: Instructed ML tutorials for graduate students (ME597) "Industrial IoT Implementation for Smart Manufacturing"

RESEARCH INTERESTS

- Robotics and Automation
- Computer vision and Artificial intelligence
- Cyber-physical system
- Smart manufacturing

MENTORING, OUTREACH, AND LEADERSHIP ACTIVITIES

- **Summer Undergraduate Research Fellowships (SURF) program**
  Purdue University – West Lafayette, IN, USA

  01. Yoonseo Lee (B.S.E in Mechanical Engineering) 2025
  - Topic: Safety-aware high-dimensional motion planning

  02. Richard Li (B.S. in Computer Science) 2024

- Topic: Optimization-based robotic grasp detection

- **Mentoring Undergraduate and Master Students**
  School of Mechanical Engineering, Purdue University – West Lafayette, IN, USA

  06. Henry Nathan Lambert (B.S.E in Mechanical Engineering) 2026 – Present
  - Topic: Robot-integrated cell-based manufacturing toward physical AI

  05. Daniel Sungwoo Bae (M.S in Mechanical Engineering) 2026 – Present
  - Topic: AI-integrated autonomous sealing robot

  04. Trevor Terrill Kates (B.S.E in Mechanical Engineering) 2026
  - Topic: Robot-integrated cell-based manufacturing toward physical AI

  03. Junho Sohn (M.S in Mechanical Engineering) 2025 – 2026
  - Topic: Interpretable AI for human motion analysis in manufacturing tasks
  (Journal submission) Sohn, J., **Lee, H.**, Lee, D., Torres, A., Neira-García, J., Zhou, Q., Abdollahi, M., Rastgaar, M., & Jun, M. (2026). HANDS: Unsupervised Segmentation for Interpretable Identification of Human Motion Primitives in Manufacturing Task. *Journal of Manufacturing Systems*.

  02. Yoonseo Lee (B.S.E in Mechanical Engineering) 2025
  - Topic: Safety-aware high-dimensional motion planning

  01. Evin Lugo Li (B.S.E in Mechanical Engineering) 2024 – 2025
  - Topic: Adaptive grasp planning for suction and two-finger grippers
  (Conference proceeding) **Lee, H.**, Choi, Y. W., Lugo, E., Lee, J., Lee, S. W., & Jun, M. B. G. (2025, June). GMGP: Generalized Model for Grasp Planning of Vacuum and Parallel Jaw Grippers. In *International Manufacturing Science and Engineering Conference* (Vol. 89022, p. V002T16A008). American Society of Mechanical Engineers.

- **Multidisciplinary Design Program (MDP)**
  University of Michigan – Ann Arbor, MI, USA
  02. Yuhang Ning (M.S in Electrical and Computer Engineering) 2022
  - Topic: Autonomous environment mapping for periodically reconfigured environments

  01. Jiwen Jia (M.S in Electrical and Computer Engineering) 2021
  - Topic: Visual servo control design for a mobile manipulator

FELLOWSHIPS, HONORS, AND AWARDS

- Best Poster Award 2026
  Manufuture Consortium 2026, West Lafayette, IN, USA
- Best Paper Award (2nd) 2026
  International Manufacturing Science and Engineering Conference (MSEC) 2026, State College, PA, USA
- Best Poster Award 2025
  U.S. – Korea Conference (UKC) 2025, Atlanta, GA, USA
- *Italian Machine Tool Technology Award* 2025
  Italian Trade Agency, Italy
- *NSF Student Travel Support Award* 2025
  ASME International Manufacturing Science and Engineering Conference (MSEC) 2025 | SME North American Manufacturing Research Conference (NAMRC) 53, Clemson University – Greenville, SC, USA
- *Young Professional NET Award* 2024 – 2025
  The Korean Federation of Science and Technology
  (Conference presentation) Sim, Y.‡*, Lee, J., **Lee, H.**, Han, C., Lee, H., & Jun, M. (2025, August 17-20). *I2DTM: Immersive and Interactive Digital Twin Metaverse Framework*

[Conference Presentation]. International Design Engineering Technical Conference & Computers and Information in Engineering Conference 2025, Anaheim, CA, United States.

(Journal publication) Jeon, J., Sim, Y., **Lee, H.,** Han, C., Yun, D., Kim, E., Nagendra, S. L., Jun, M., Kim, Y., Lee, S., & Lee, J.* (2025). ChatCNC: Conversational Machine Monitoring via Large Language Model and Real-Time Data Retrieval Augmented Generation. *Journal of Manufacturing Systems*, *79*, 504-514.

- *Best Poster Award* 2024
  DigiTwin 2024 for Manufacturing, Sustainability, Safety, and Resilience
- *Winkelman Fellowship* 2023 – 2024
  Purdue University – West Lafayette, IN, USA
- *DSCD and AACC Student Travel Awards* 2022
  The 2022 Modeling Estimation and Control Conference (MECC), Jersey City, NJ, USA.
- *Multidisciplinary Design Program (MDP) Summer Fellowship* 2021
  University of Michigan – Ann Arbor, MI, USA

JOURNAL PUBLICATIONS

‡: Equal contribution ***: Corresponding author**

11. Sim, Y.*, Kim, E., Sohn, J., Barr, J., **Lee, H.**, Beach, B., Jun, M. & Shakouri, A. (2026). Generative Latent Space Modeling of Stamping Sound for Anomaly Detection and Predictive Maintenance. *Journal of Manufacturing Science and Engineering*, doi: https://doi.org/10.1115/1.4072576 (2nd Best Paper Award)

10. **Lee, H.**, Sim, Y., Yoon, B., Lee, H., Park, H., Jang, J., Kim, C., & Jun, M.* (2026). Toward provably safe AI-driven perception and motion planning for robotic manufacturing: A review. *Journal of Manufacturing Systems*, doi: 10.1016/j.jmsy.2026.07.014 (#1 in Operations Research & Management Science OOIR 2026)

09. **Lee, H.**, Han, C., Gabor, T., Sim, Y., Akin, S., Jun, M., & Lee, J.* (2026). Cold spray-based secure and unique product identification with neural encoding: A full-stack framework for scalable authentication in manufacturing. *Journal of Intelligent Manufacturing*, doi: https://doi.org/10.1007/s10845-026-02864-w

08. Sim, Y., Kim, E.*, **Lee, H.**, Sohn, J., & Jun, M. (2025). Structured Sound Data Augmentation Based on Spectral Analysis for Machine Tool Operational State Monitoring. *The International Journal of Advanced Manufacturing Technology*, *141*, 1705-1719. doi: https://doi.org/10.1007/s00170-025-16797-w

07. Kim, E., **Lee, H.**, Sim, Y., Lee, J., & Jun, M.* (2025). Overcoming Sparse Run-to-Failure Data Challenges in Manufacturing: A Contrastive Mixer Framework for Remaining Useful Life Prediction. *CIRP Annals – Manufacturing Technology*, *74*, 579-583. doi: https://doi.org/10.1016/j.cirp.2025.04.082

06. Han, C., Gabor, T., **Lee, H.**, Lee, J., Akin, S., & Jun, M.* (2025). Pulsed Cold Spray System for Physical Unclonable Function Generation. *Procedia CIRP*, *137*, 176-180. doi: https://doi.org/10.1016/j.procir.2025.02.269

05. Jeon, J., Sim, Y., **Lee, H.**, Han, C., Yun, D., Kim, E., Nagendra, S. L., Jun, M., Kim, Y., Lee, S., & Lee, J.* (2025). ChatCNC: Conversational Machine Monitoring via Large Language Model and Real-Time Data Retrieval Augmented Generation. *Journal of Manufacturing Systems*, *79*, 504-514. doi: https://doi.org/10.1016/j.jmsy.2025.01.01 (#1 in Operations Research & Management Science OOIR 2025)

04. **Lee, H.***, Toner, T., Tilbury, D., & Barton, K. (2025). GraspMixer: Hybrid of Contact Surface Sampling and Grasp Feature Mixing for Grasp Synthesis. *IEEE Transactions on Automation Science and Engineering*, *22*, 10920- 10936. doi: 10.1109/TASE.2025.3530795 (#12 in Automation & Control System OOIR 2025)

03. Lee, J., Akin, S.*, Sim, Y., **Lee, H.**, Kim, E., Nam, J., Song, K., & Jun, M.* (2024). A stethoscope-guided interpretable deep learning framework for powder flow diagnosis in cold spray additive manufacturing. *Manufacturing Letters*, *41*, 1515-1525. doi: https://doi.org/10.1016/j.mfglet.2024.09.178

02. Han, C., Lee, J., **Lee, H.**, Sim, Y., Jeon, J., & Jun, M.* (2024). Zero-shot Autonomous Robot Manipulation via Natural Language. *Manufacturing Letters*, *42*, 16-20. doi: https://doi.org/10.1016/j.mfglet.2024.09.067

01. **Lee, H.***, Toner, T., Tilbury, D., & Barton, K. (2022). Multi-sensor aided deep pose tracking. *IFAC-PapersOnLine*, *55*(37), 326-332. doi: https://doi.org/10.1016/j.ifacol.2022.11.205

JOURNAL IN PROCESS

‡: Equal contribution ***: Corresponding author**

12. (In preparation) **Lee, H.**, Sim, Y., Lee, H., Lee, K., Jun, M., & Lee, J.* (2026). BRIDGES: Bi-directional Robotic Integration with Digital-twin for Generalized, Efficient, and Safe Automation in Manufacturing.

11. (In preparation) Sim, Y., **Lee, H.**, Lee, H., Paek, S., Lee, J., Yoon, B., & Jun, M.* (2026). MMRL: A Multi-Modal Avatar based Remote Interaction and Localization Framework for Digital Twin Cross-Validation in Smart Manufacturing.

10. (In preparation) Lee, H., Sim, Y., Yoon, B., Kim, C., & Jun, M.* (2026). Silhouette-guided Robust Object 6D Pose Estimation and Certification.

09. (Under review, submitted in Oct. 2026) Lee, H., Sim, Y., **Lee, H.**, Yoon, B., Lee, J., Garcia, J. E. N., Rastgaar, M., Jang, J., & Jun, M.* (2026). High-Precision 3D Trajectory Estimation Using Wire Encoder based Module with String Dynamic Model Compensation. *Manufacturing Letters*.

08. (Under review, submitted in Oct. 2026) Sim, Y., **Lee, H.**, Kim, E., Shakouri, A., & Jun, M.* (2026). Acoustic Fingerprinting for Online Condition Monitoring in Progressive Stamping. *Journal of Manufacturing Processes*.

07. (Under review, submitted in Oct. 2026) **Lee, H.**, Sim, Y., Yoon, B., Kim, C., & Jun, M.* (2026). Cuboid Superposition for Rapid Scene-Adaptive C-Space Modeling and Real-Time Path Planning. *IEEE Transaction on Robotics*.

06. (Under review, submitted in Sep. 2026) Sim, Y., **Lee, H.**, Tanveer, A., Ha, H., Lee, H., Lee, Y., Kim, E., Jung, B., & Jun, M.* (2026). TinyML Dielectric Spark Detection via Physics-Based Acoustic Event Synthesis. *Engineering Applications of Artificial Intelligence*.

05. (Under revision, submitted in Mar. 2026) **Lee, H.**, Sim, Y., Lee, H., Yoon, B., Kim, C., & Jun, M.* (2025). PRISM: Progressive Refinement via Informed Sampling for Multi-belief Object Pose Validation. *Robotics and Computer-Integrated Manufacturing*.

04. (Under review, submitted in Jan. 2026) Sohn, J., **Lee, H.**, Lee, D., Torres, A., Neira-García, J., Zhou, Q., Abdollahi, M., Rastgaar, M., & Jun, M.* (2026). HANDS: Unsupervised Segmentation for Interpretable Identification of Human Motion Primitives in Manufacturing Task. *Engineering Applications of Artificial Intelligence*.

03. (Under review, submitted in Jan. 2026) Jun, M.*, Sim, Y., Kim, E., **Lee, H.** (2026). Open-world condition recognition for discrete manufacturing. *CIRP Journal of Manufacturing Science and Technology*.

02. (Under revision, submitted in Dec. 2025) Nam, J., **Lee, H.**, Han, C., Athinarayanan, R., Jun, M., & J, Lee.* (2025). C³AM: Conversational CAD/CAM Automation via Multimodal Large Language Model Agents. *Journal of Intelligent Manufacturing*.

01. (In second round of review) **Lee, H.**, Sim, Y., Han, C., Jun, M., & Lee, J.* (2025). GRAVITON: Generalized Resampling Algorithm for Visual Inference in Target Object Localization. *Computers in Industry*.

PEER-REVIEWED CONFERENCE PROCEEDINGS

‡: Equal contribution ***: Corresponding author**

15. (In preparation) Lee, H., Sim, Y., **Lee, H.**, Yoon, B., Lee, J., Garcia, J. E. N., Rastgaar, M., Jang, J., & Jun, M.* (2027, November). High-Precision 3D Trajectory Estimation Using Wire Encoder based Module with String Dynamic Model Compensation. In 54th North American Manufacturing Research Conference. American Society of Mechanical Engineers.

14. (In preparation) **Lee, H.***, Yoon, B., Sim, Y., Kim, C., & Jun, M. B. G. (2027, October). *Autonomous Cnc Chip Removal Through Pneumatic Non-Prehensile Manipulation: Toward Process Assistant Robots*. In 2027 *International Manufacturing Science and Engineering Conference*, Corvallis, OR, USA.

13. (In preparation) **Bae, D.***, Jun, M., **Lee, H.**, Sim, Y., & Lee, H. (2027, October). *Development and Evaluation of a Robotic Silicone Dispensing System*. In 2027 *International Manufacturing Science and Engineering Conference*, Corvallis, OR, USA.

12. (In preparation) Sim, Y., Kim, E., **Lee, H.**, Yoon, B., Jun, Y., Lee, Y., & Jun, M. B. G. (2027, October). *Manucodec: Manufacturing Aware Neural Audio Encoding for Scalable Sound Acquisition*. In 2027 *International Manufacturing Science and Engineering Conference*, Corvallis, OR, USA.

11. (Accepted in Apr. 2026) Jun, M.*, Sim, Y., **Lee, H.**, Kim, E., & Shakouri, A. (2026, October 11-16). *Acoustic Fingerprinting for Online Condition Monitoring in Progressive Stamping*. International Conference on the Technology of Plasticity 2026, Jeju Island, Republic of Korea.

10. (Accepted in Jun. 2026) Yoon, B., **Lee, H.**, Sim, Y., & Jun, M.* **(**2026, September). LiDAR-Integrated Coarse-to-Fine Optimization for Geometrically Consistent Triangle Splatting from Mobile Robots. In *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE.

09. Lee, H., Sim, Y., Yoon, B., **Lee, H.**, & Jun, M.* (2026, June). Development of Scalable FSR Sensor Array-based Human Footprint Acquisition Module for Manufacturing Automation. In *Excellence in Manufacturing and Operations (XMO) Industrial Seminar 2026*. doi: 10.5703/1288284318684

08. Sim, Y., Lee, Y., Yother, E., Lee, D., **Lee, H.**, Yoon, B., & Jun, M.* (2026, June). Real-time Two-stage Machine Sound Monitoring for Wire-EDM State Recognition. In *Excellence in Manufacturing and Operations (XMO) Industrial Seminar 2026*. doi: 10.5703/1288284318685

07. Bae, D., Han, C., Lee, H., **Lee, H.**, & Jun, M.* (2026, June). RGB-D Object Pose Estimation via SAM2 Segmentation and Coarse-to-Fine Point Cloud Registration. In *Excellence in Manufacturing and Operations (XMO) Industrial Seminar 2026*. doi: 10.5703/1288284318682

06. (Accepted in Jan. 2026) Nam, J.*, **Lee, H.**, Han, C., Athinarayanan, R., Jun, M., & J, Lee. (2026, June). Gen-Cad/cam: Generative and Interactive Cad/cam Framework for Simulation-Informed Adaptive Maintenance. In *International Manufacturing Science and Engineering Conference*. American Society of Mechanical Engineers.

05. (Under review, submitted in Jan. 2026) Han, C., Sim, Y., Jung, H., Lee, J., **Lee, H.**, Kang, Y., Woo, S., Kim, G., Park, H., & Jun, M.* (2026, July). IMPACT: Industrial Machine Perception via Acoustic Cognitive Transformer. In *Knowledge Discovery and Data Mining 2026 (KDD)*. ACM.

04. **Lee, H.***, Sim, Y., Han, C., Lee, J., Bera, A., Kim, C., & Jun, M. B. G. (2025, October). EASEIR: Efficient and Adaptive Safe-set Estimation via Implicit Representation for High-dimensional Motion Planning. In *2025*

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (pp. 11661-11668). IEEE. doi: 10.1109/IROS60139.2025.11246375

03. Sim, Y.*, Kim, E., Lee, J., **Lee, H.**, Yun, D., Myeong, N., Hwang, Y., Park, H., & Jun, M. B. G. (2025, June). Sound Data Augmentation Using Frequency Response Superposition for Machine Tool Component Running State Recognition. In *International Manufacturing Science and Engineering Conference* (Vol. 89022, p. V002T16A006). American Society of Mechanical Engineers. doi: https://doi.org/10.1115/MSEC2025-155613

02. **Lee, H.***, Choi, Y. W., Lugo, E., Lee, J., Lee, S. W., & Jun, M. B. G. (2025, June). GMGP: Generalized Model for Grasp Planning of Vacuum and Parallel Jaw Grippers. In *International Manufacturing Science and Engineering Conference* (Vol. 89022, p. V002T16A008). American Society of Mechanical Engineers. doi: https://doi.org/10.1115/MSEC2025-155673 (NSF Student Travel Support Award)

01. Han, C.*, **Lee, H.**, Lee, J., Jun, M. B. G., & Yun, H. (2024, June). A Study on Digital Twin for Autonomous Robotic Chip Removal Deep Learning Framework on CNC Machine. In *International Manufacturing Science and Engineering Conference* (Vol. 88100, p. V001T02A003). American Society of Mechanical Engineers. doi: https://doi.org/10.1115/MSEC2024-124383

CONFERENCE PRESENTATIONS

‡: Presenter ***: Corresponding author**

14. Jun, M.‡*, **Lee, H.**, Sim, Y., Lee, H., Yoon, B., & Bae, D. (2026, June 28- July 3). *Leveraging a Vision Foundation Model for Adaptable and Robust Object 3D Localization and Manipulation under Real Shop-Floor Environments*. International Conference on Precision Engineering and Sustainable Manufacturing 2026, Singapore.

13. Nam, J.*, **Lee, H.**, Han, C., Athinarayanan, R., Jun, M., & J, Lee. (2026, June 14-18). *Gen-Cad/cam: Generative and Interactive Cad/cam Framework for Simulation-Informed Adaptive Maintenance*. 2026 International Manufacturing Science and Engineering Conference, State College, PA, United States.

12. Sim, Y.*, Kim, E., Sohn, J., Barr, J., **Lee, H.**, Beach, B., Jun, M. & Shakouri, A. (2026, June 14-18). *Generative Latent Space Modeling of Stamping Sound for Anomaly Detection and Predictive Maintenance*. 2026 International Manufacturing Science and Engineering Conference, State College, PA, United States.

11. **Lee, H.**‡*, Sim, Y., Lee, H., Yoon, B., Kim, C., & Jun, M. (2026, January 29-30). *PRISM: Progressive Refinement via Informed Sampling for Multi-belief Object Pose Estimation*. 2026 Institute for Control, Optimization and Network Student Research Conference, West Lafayette, IN, United States.

10. **Lee, H.**‡*, Sim, Y., Han, C., Lee, J., Bera, A., Kim, C., & Jun, M. B. G. (2025, October 19-25). *EASEIR: Efficient and Adaptive Safe-set Estimation via Implicit Representation for High-dimensional Motion Planning*. 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems, Hangzhou, China.

09. Kim, E., **Lee, H.**, Sim, Y., Lee, J., & Jun, M.‡* (2025, August 17-23). *Overcoming Sparse Run-to-Failure Data Challenges in Manufacturing: A Contrastive Mixer Framework for Remaining Useful Life Prediction*. 74th CIRP Annals – Manufacturing Technology, Stockholm, Sweden.

08. Sim, Y.‡*, Lee, J., **Lee, H.**, Han, C., Lee, H., & Jun, M. (2025, August 17-20). *I2DTM: Immersive and Interactive Digital Twin Metaverse Framework*. International Design Engineering Technical Conference & Computers and Information in Engineering Conference 2025, Anaheim, CA, United States.

07. **Lee, H.**, Sim, Y., Han, C., Lee, J., & Jun, M.‡* (2025, July 6-11). *ARMPASS: Autonomous Robot Motion Planning with Adaptive Safe Set*. International Conference on Precision Engineering and Sustainable Manufacturing 2025, Chiang Mai, Thailand.

06. Sim, Y.‡*, Kim, E., Lee, J., **Lee, H.**, Yun, D., Myeong, N., Hwang, Y., Park, H., & Jun, M. B. G. (2025, June 23-27). *Sound Data Augmentation Using Frequency Response Superposition for Machine Tool Component Running State Recognition*. International Manufacturing Science and Engineering Conference 2025, Greenville, SC, USA.

05. **Lee, H.‡***, Choi, Y. W., Lugo, E., Lee, J., Lee, S. W., & Jun, M. B. G. (2025, June 23-27). *GMGP: Generalized Model for Grasp Planning of Vacuum and Parallel Jaw Grippers*. International Manufacturing Science and Engineering Conference 2025, Greenville, SC, USA.

04. Han, C.‡, Gabor, T., **Lee, H.**, Lee, J., Akin, S., & Jun, M.* (2025, June 1-4). *Pulsed Cold Spray System for Physical Unclonable Function Generation*. CIRP International Symposium on Electro Physical and Chemical Machining 2025, Vancouver, BC, Canada.

03. Jun, M.‡*, Lee, J., Kim, E., Sim, Y., Han, C., & **Lee, H.** (2024, July 7-12). *Generalized and Generative AI for Smart Manufacturing*. International Conference on Precision Engineering and Sustainable Manufacturing 2024, Busan, Republic of Korea.

02. Han, C.‡*, **Lee, H.**, Lee, J., Jun, M. B. G., & Yun, H. (2024, June 17-21). *A Study on Digital Twin for Autonomous Robotic Chip Removal Deep Learning Framework on CNC Machine*. International Manufacturing Science and Engineering Conference 2024. Knoxville, TN, USA.

01. **Lee, H.‡***, Toner, T., Tilbury, D., & Barton, K. (2022, October 2-5). *Multi-sensor aided deep pose tracking*. Modeling, Estimation and Control Conference 2022, Jersey City, NJ, United States. (DSCD and AACC Student Travel Awards)

POSTER SESSIONS

12. **Lee, H.**, Sim, Y., Lee, H., Yoon, B., Kim, C., & Jun, M.* (2026, September 28-30). *Leveraging a Vision Foundation Model for Adaptable and Robust Object Localization*. Manufuture Consortium 2026, West Lafayette, IN, United States.

11. Nam, S., **Lee, H.**, Sim, Y., Athinarayanan, R., Jun, M., & Lee, J.* (2026, September 28-30). *Gen-CAD/CAM: Generative and Interactive CAD/CAM Framework using Large Language Models*. Manufuture Consortium 2026, West Lafayette, IN, United States.

10. Sim, Y., Kim, E., Beach, B., **Lee, H.**, Jun, M. & Shakouri, A.* (2026, September 28-30). *Generative AI for Abnormal Stamping Sound Detection and Predictive Maintenance*. Manufuture Consortium 2026, West Lafayette, IN, United States. (Best Poster Award)

09. Lee, H., Sim, Y., Yoon, B., **Lee, H.**, & Jun, M.* (2026, June 2-3). *Development of Scalable FSR Sensor Array-based Human Footprint Acquisition Module for Manufacturing Automation*. Excellence in Manufacturing and Operations (XMO) Industrial Seminar 2026, West Lafayette, IN, United States.

08. Sim, Y., Lee, Y., Yother, E., Lee, D., **Lee, H.**, Yoon, B., & Jun, M.* (2026, June 2-3). *Real-time Two-stage Machine Sound Monitoring for Wire-EDM State Recognition*. Excellence in Manufacturing and Operations (XMO) Industrial Seminar 2026, West Lafayette, IN, United States.

07. Bae, D., Han, C., Lee, H., **Lee, H.**, & Jun, M.* (2026, June 2-3). *RGB-D Object Pose Estimation via SAM2 Segmentation and Coarse-to-Fine Point Cloud Registration*. Excellence in Manufacturing and Operations (XMO) Industrial Seminar 2026, West Lafayette, IN, United States.

06. Han, C., Sim, Y., Lee, J., **Lee, H.**, & Jun, M. (2025, August 5-9). Industrial Audio Foundation for Agile Machine Listening. U.S. – Korea Conference 2025, Atlanta, GA, United States. (Best Poster Award)

05. Sim, Y., Lee, J., **Lee, H.**, Yun, D., Myeong, N., Hwang, Y., Park, H., Jun, M., & Kim, E. (2024, December 2). *Sound Data Augmentation Using Frequency Response Superposition for Machine Tool State Recognition*. Manufacturing And Materials Research Laboratories Symposium 2024, West Lafayette, IN, United States.

04. Lee, J., Jeon, J., Sim, Y., **Lee, H.**, Han, C., Yun, D., & Jun, M. (2024, October 15-16). *Generative AI Based Real-Time Human-Data Interaction for Smart Manufacturing*. Digital Twin for Manufacturing Sustainability, Safety, and Resilience 2024, West Lafayette, IN, United States. (Best Poster Award)

03. **Lee, H.**, Sim, Y., Lee, J., Han, C., & Jun, M. (2024, October 15-16). *Digital Twin & Vision Guided Autonomous Robotic Path Planning*. Digital Twin for Manufacturing Sustainability, Safety, and Resilience 2024, West Lafayette, IN, United States.

02. Lee, J., Kim, E., Han, C., **Lee, H.**, & Sim, Y. (2024, March 24-26). *Generalized and Generative Smart Manufacturing*. North Academy of Engineering Regional Meeting 2024, West Lafayette, IN, United States.

01. Han, C., **Lee, H.**, & Jun, M. (2024, May 24-26). *Robotics in Smart Manufacturing*. Congressional Artificial Intelligence Caucus 2024, Washington, D.C., United States.

PATENTS

03. Lee, H., Sim, Y., **Lee, H.**, Yoon, B., Lee, J., Garcia, J. E. N., Rastgaar, M., Jang, J., & Jun, M.* (2027, June 1). *High-Precision 3D Trajectory Estimation Using Wire Encoder based Module with String Dynamic Model Compensation* (Track Code 2027-JUN-72011). U.S. Patent and Trademark Office.

02. Sim, Y., **Lee, H.**, Tanveer, A., Ha, H., Lee, H., Lee, Y., Kim, E., Jung, B., & Jun, M.* (2026, June 1). *TinyML Dielectric Spark Detection via Physics-Based Acoustic Event Synthesis* (Track Code 2027-JUN-72011). U.S. Patent and Trademark Office.

01. Jun, M., Lee, J., **Lee, H.**, Akin, S., Han, C., Gabor, T., & Sim, Y. (2025, September 18). *Cold spray-enabled physically unclonable identifier and its spectral authentication via implicit neural representation* (U.S. Patent Application No. 64/033,490). U.S. Patent and Trademark Office.

PROFESSIONAL SOCIETY MEMBERSHIPS & SERVICES

- Reviewer of the IEEE Transactions on Automation Science and Engineering — 2025 – Present
- Reviewer of the IEEE/RSJ International Conference on Intelligent Robots and Systems — 2025 – Present
- Reviewer of the ASME Manufacturing Science and Engineering Conference — 2025 – Present
- Member of the Institute of Electrical and Electronics Engineers (IEEE) — 2025 – Present
- Member of the American Society of Mechanical Engineers (ASME) — 2025 – Present
- Member of the Society of Manufacturing Engineers (SME) — 2025 – Present
- Graduate Association for Manufacturing Excellence / Treasurer — 2025 – Present
- Korean-American Scientists and Engineers Association (KSEA) / Member — 2024 – Present

INDUSTRIAL PROJECTS

- **Genesem, Inc.** — 2026 – Present
  - Development of a CAD-based part localization in a semiconductor equipment for part replacement.

- **Hitachi America, Ltd.** — 2024 – Present
  - Developed a multi-modal feature extraction method to fuse data from a glove sensor and a camera.
  - Extended the feature extraction module to perform motion analysis and task segmentation of human operators during manufacturing processes.

- **BCI Solution, Inc** 2024 – 2025
  - Implemented a digital twin of FANUC CRX-10iA/L using MuJoCo physics simulator and Python.
  - Established bi-directional communication among FANUC CRX-10iA, its digital twin, and CESMII platform for automation and verification of core-making operations.

- **Korea Institute of Machinery and Materials (KIMM)** 2024 – Present
  - Developed a generalized sampling algorithm to extend a cutting edge object segmentation AI to 6-DOF object pose estimation method for machine tending.
  - Developed a hybrid computer vision pipeline to localize metal debris inside CNC machine that must be removed.

- **BCI Solution, Inc** 2023 – Present
  - Built a server-client system using Python and KAREL to communicate FANUC controller via external PC.
  - Integrated 6-DOF object pose estimation algorithm with FANUC controller via the server-client system for tool-path adaptation

- **Terra Drive Systems (TDS)** 2023 – 2023
  - Installed a stethoscope-based sound sensor and a mini PC to establish a data pipeline.
  - Developed a sound recognition AI model to monitor status of Makino a81 CNC machine.
  - Generated a web-based dashboard for intuitive data analysis and visualization.